Yellowtail flounder (*Limanda ferruginea*)

Summary Scores

- **Sensitivity:** Low - Moderate
- **Exposure:** Moderate - High
- **Vulnerability:** Low - Moderate

Sensitivity

Sensitivity Attributes	Range Score	Range LH DQ	Range Tally	Georges Bank Score	Georges Bank LH DQ	Georges Bank Tally	Gulf of Maine/Cape Cod Score	Gulf of Maine/Cape Cod LH DQ	Gulf of Maine/Cape Cod Tally	Southern New England/Mid-Atlantic Score	Southern New England/Mid-Atlantic LH DQ	Southern New England/Mid-Atlantic Tally
Adult Mobility	2.37	2.36		2.5	2.36		2.32	2.36		2.32	2.36	
Complexity in Reproductive Strategy	1.76	1.97		1.76	2.04		1.76	2.04		1.76	1.84	
Dispersal of Early Life Stages	1.32	2.74		1.32	2.74		1.32	2.74		1.32	2.74	
Egg and Larval Survival and Settlement Requirements	2.57	2.47		2.6	2.5		2.48	2.4		2.64	2.5	
Historical Variability in Temperature	1.57	2.8		1.56	2.8		1.6	2.8		1.56	2.8	
Juvenile Habitat Requirements	1.2	2.5		1.2	2.5		1.2	2.5		1.2	2.5	
Other Stressors	2.29	2.61		2.28	2.6		2.4	2.64		2.2	2.6	
Population Growth Rate	2.11	2.72		2.08	2.72		2.04	2.72		2.2	2.72	
Prey Specificity	1.52	2.7		1.52	2.7		1.52	2.7		1.52	2.7	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.4	2.58		1.4	2.58		1.4	2.58		1.4	2.58	
Spawning Cycle	2.55	2.48		2.56	2.48		2.36	2.48		2.72	2.48	
Stock Size/Status	3.44	2.46		3.56	2.21		2.8	2.75		3.96	2.43	
Total Sensitivity | Certainty	**Moderate (2) | 0.84**			**Moderate (2) | 0.88**			**Low (1) | 0.56**			**Moderate (2) | 0.88**		

Exposure

Exposure Factors	Range Mean	Range SD	Range Tally	Georges Bank Mean	Georges Bank SD	Georges Bank Tally	Gulf of Maine/Cape Cod Mean	Gulf of Maine/Cape Cod SD	Gulf of Maine/Cape Cod Tally	Southern New England/Mid-Atlantic Mean	Southern New England/Mid-Atlantic SD	Southern New England/Mid-Atlantic Tally
Bottom Temperature	2.8	1.01		2.49	0.95		3	1.02		3.07	0.85	
Bottom Oxygen	3.09	0.89		2.72	0.80		3.46	0.80		3.22	0.81	
Bottom Salinity	2.83	0.91		3.44	0.66		1.97	0.90		3.08	0.43	
Bottom Aragonite Solubility	1.12	0.46		1.29	0.76		1.04	0.23		1.02	0.17	
Net primary production	1	0.03		1	0.01		1	0.03		1	0.02	
Downward Flux of Particulate Organic Carbon	1.2	0.26		1.28	0.25		1.26	0.31		1.06	0.14	
Total Exposure	**Moderate (2.34)**	**0.89**		**Moderate (2.21)**	**0.99**		**Moderate (2.45)**	**0.83**		**High (2.5)**	**0.77**	

Vulnerability

Vulnerability	Range Mean	Range SD	Range Tally	Georges Bank Mean	Georges Bank SD	Georges Bank Tally	Gulf of Maine/Cape Cod Mean	Gulf of Maine/Cape Cod SD	Gulf of Maine/Cape Cod Tally	Southern New England/Mid-Atlantic Mean	Southern New England/Mid-Atlantic SD	Southern New England/Mid-Atlantic Tally
Total Vulnerability	**Moderate (3.96)**	**1.88**		**Moderate (4.42)**	**1.98**		**Low (2.5)**	**0.95**		**Moderate (5)**	**1.55**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Yellowtail flounder. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 6 and Figure 5). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Low	Low - Moderate	None - Increase
Exposure	High	Moderate - High	Decrease - None
Vulnerability	Low	Low - Moderate	None - Increase
Directional Effect	Negative	Neutral - Negative	Increase - None

Stocks

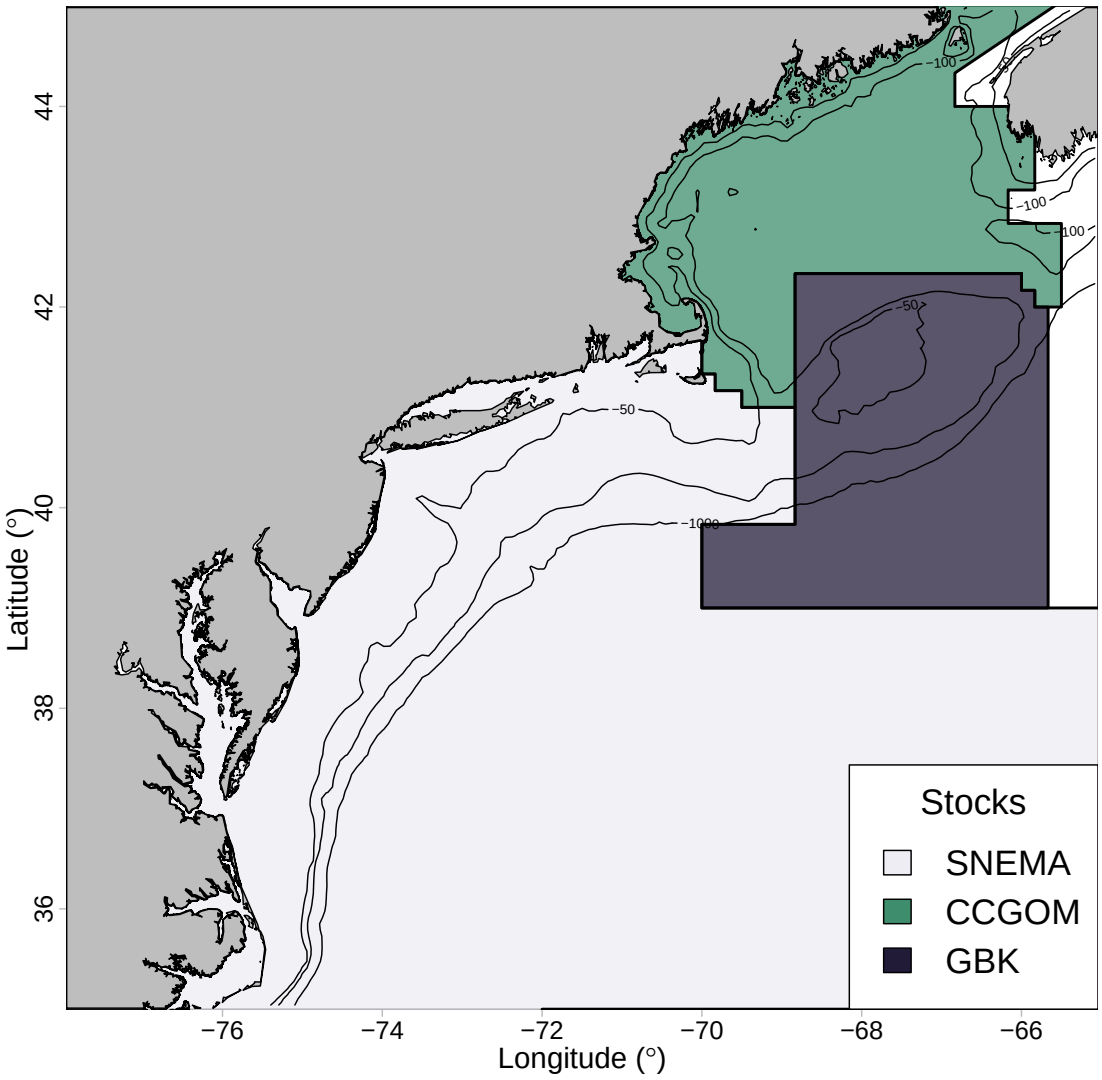


Figure 2: Yellowtail flounder Stocks along the Northeast Shelf

Score Descriptions:

Sensitivity:

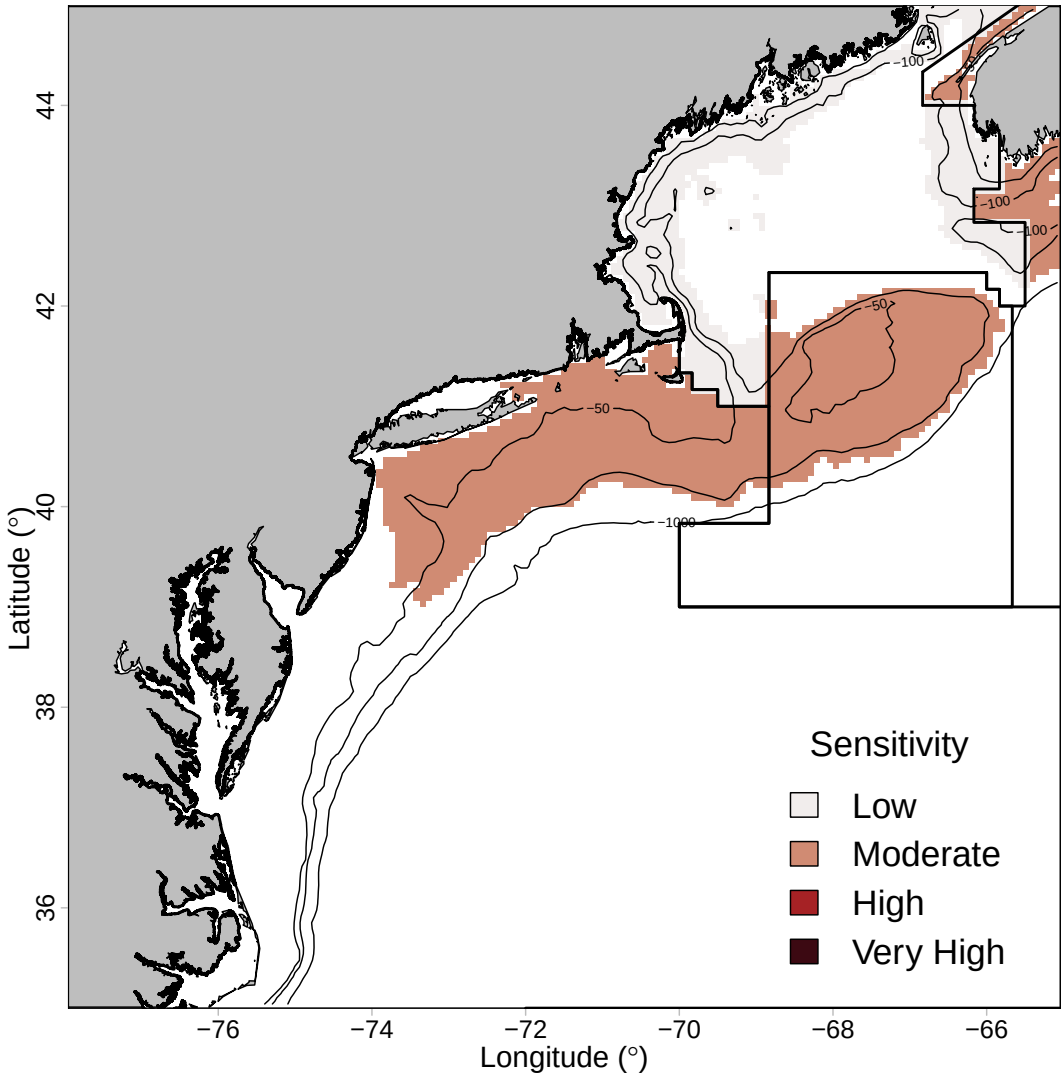


Figure 3: Yellowtail flounder Sensitivity on the Northeast Shelf

The **Southern New England/Mid-Atlantic** and **Georges Bank** stocks of Yellowtail flounder both received **Moderate** sensitivity scores. Both scores had 88% certainty. These scores were driven by Stock Size (4 and 3.6), Spawning Cycle (2.7 and 2.6) and Egg and Larval Survival and Settlement Requirements (2.6). Adult Mobility (2.5) also played a role in the score for the Georges Bank stock (Figure 1).

The **Gulf of Maine/Cape Cod** stock of Yellowtail flounder received a **Low** sensitivity score, with 56% certainty. For this stock, only the Stock Size (2.8) attribute had a score greater than 2.5 (Figure 1).

Spawning occurs from mid-February to July on Georges Bank and in nearshore habitats in Southern New England and the Mid-Atlantic (Martin and Drewry 1978). Larvae feed on zooplankton prey and zooplankton abundance has been linked to recruitment success. Although it is also worth noting that studies directly testing for mismatches did not find evidence of mismatches between larvae and zooplankton

overlap and recruitment (J. Kittel 2025) (Johnson et al. 1999) McManus and Richardson 2023). Larvae perform vertical migrations to help increase retention in and around nursery locations. Recruitment has also been correlated with circulation patterns and the Mid Atlantic Cold Pool. Both the Georges Bank and Southern New England stocks are under rebuilding plans (Northeast Fisheries Science Center (NEFSC) 2025b) (Northeast Fisheries Science Center (NEFSC) 2025c).

In the 2016 Climate Vulnerability Assessment, yellowtail flounder also received a Low sensitivity score. Similar to the Gulf of Maine/Cape Cod stock, the only attribute that scored above 2.5 in the 2016 Assessment was Stock Status (Hare et al. 2016).

Exposure:

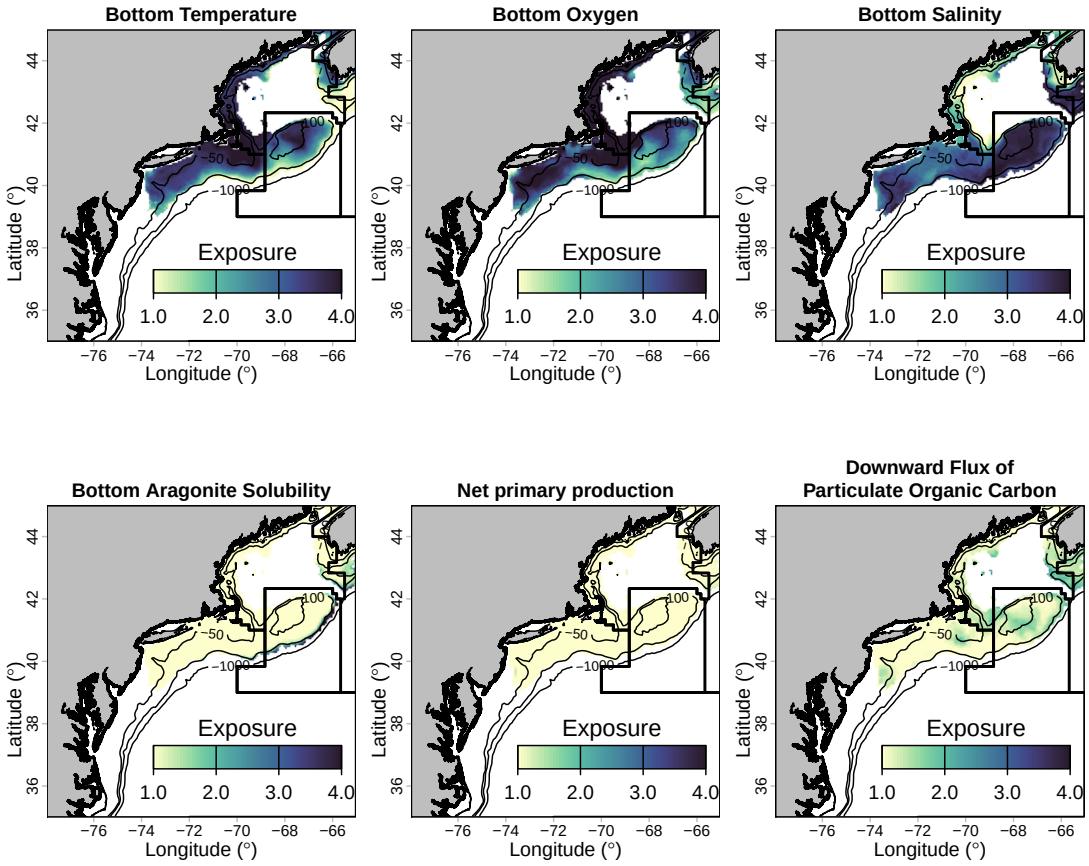


Figure 4: Maps of each Exposure Factor for Yellowtail flounder. Exposure was calculated between 2014-2023 and 2025-2035

Mean exposure for Yellowtail flounder across its range was **Moderate** (2.34 +/- 0.89). Mean scores were also **Moderate** within the **Georges Bank** and **Gulf of Maine/Cape Cod** stocks (2.21 +/- 0.99 and 2.45 +/- 0.83, respectively). Average exposure was **High** for the **Southern New England/Mid-Atlantic** stock (2.5 +/- 0.77) (Figure 1). Exposure was Very High over Georges Bank and High throughout most of Southern New England and along the coast of the Gulf of Maine (Figure 5). These patterns were driven by high bottom temperature, oxygen, and salinity exposure (Figure 4).

In the previous Assessment, yellowtail flounder received a High Exposure score (Hare et al. 2016).

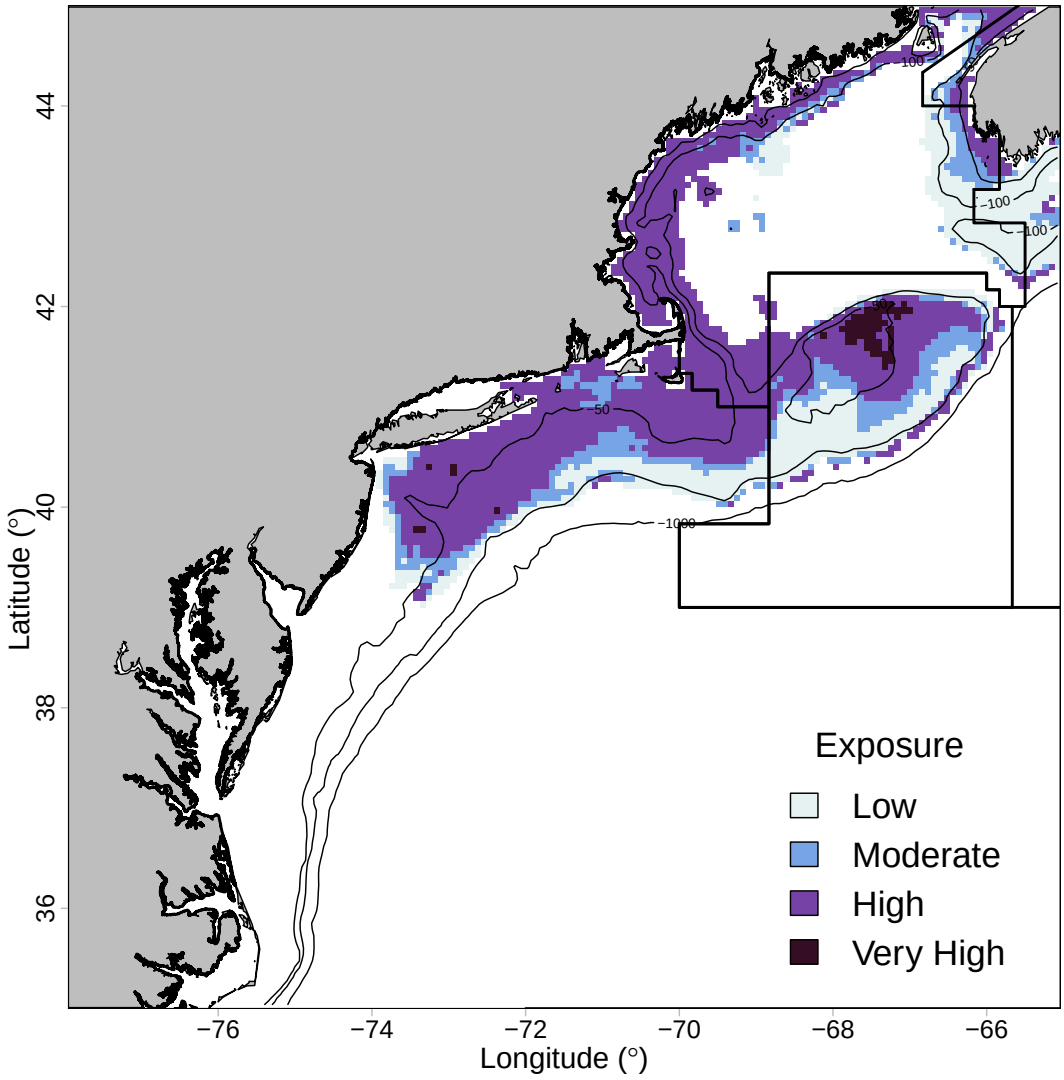


Figure 5: Total Exposure on the Northeast Shelf for Yellowtail flounder. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 4)

Vulnerability:

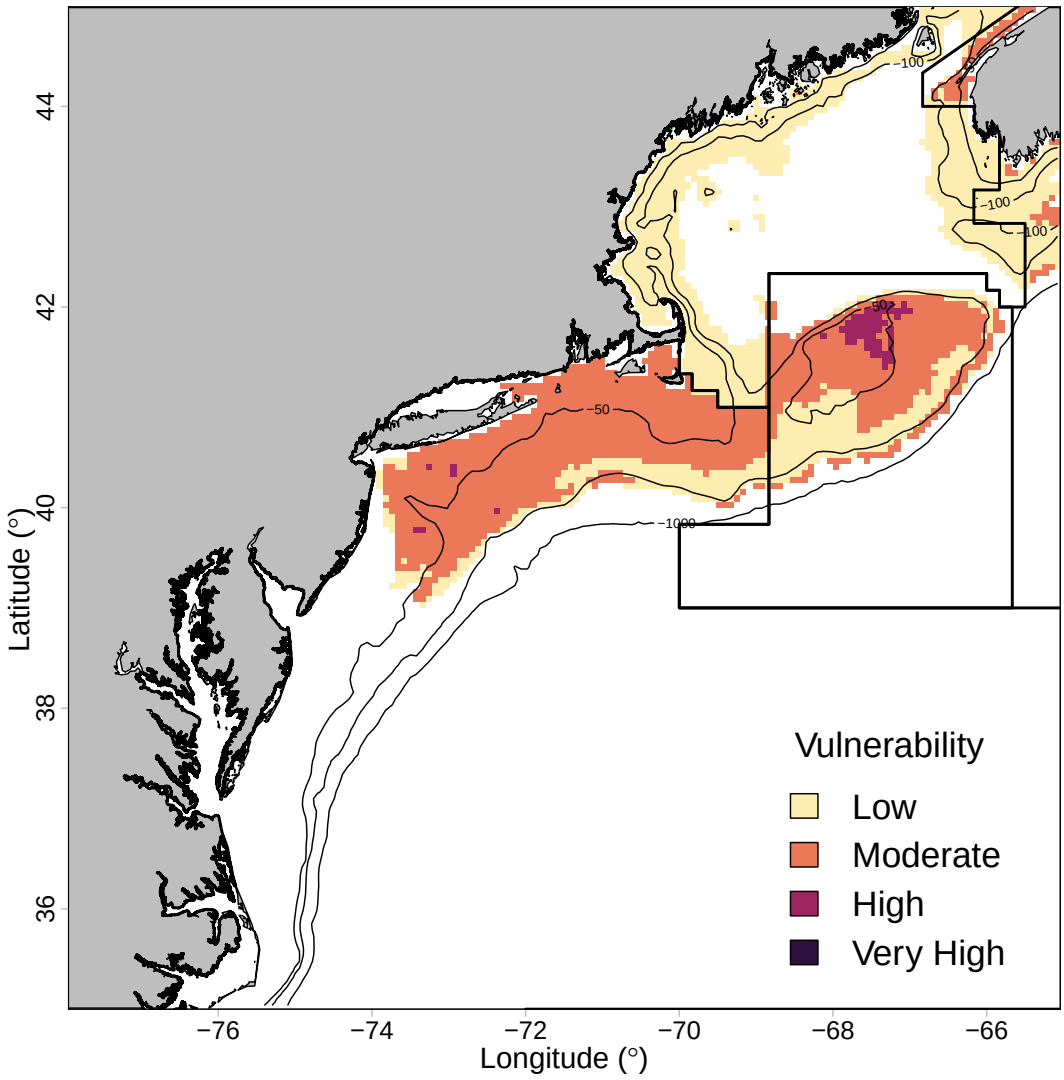


Figure 6: Vulnerability on the Northeast Shelf for Yellowtail flounder. Vulnerability is the product of sensitivity and exposure (see Figure 5 and Figure 3)

Yellowtail flounder received a mean Vulnerability score of **Moderate** (3.96 +/- 1.88) (Figure 1). Scores were **Moderate** in the **Georges Bank** and **Southern New England/Mid-Atlantic** stocks (4.42 +/- 1.98 and 5 +/- 1.55, respectively). Mean Vulnerability was **Low** in the **Gulf of Maine/Cape Cod** stock due to Low Sensitivity (2.5 +/- 0.95) (Figure 1). Vulnerability was High over portions of Georges Bank, and Moderate over the Bank and in Southern New England (Figure 6). These scores did not change when exposure was calculated using only factors important to yellowtail distribution (Figure 1).

In the previous Assessment, yellowtail flounder received a Low Vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

Between 93 - 97% of data quality scores were greater than or equal to 2 across both stocks, indicating excellent data quality. Complexity in Reproductive Strategy was identified as a data gap (mean attribute data quality score < 2) for the Southern New England/Mid-Atlantic stock (Figure 1).

Model Confidence Score:

The yellowtail distribution model received a 2.4 (+/- 0.2) mean model confidence score, indicating moderate confidence in the model. Experts noted that the model overpredicted distributions over Georges Bank, but otherwise accurately predicted distributions. The ensemble model AUC score was 0.88, indicating good model performance.

Additional Metrics

Distribution Shifts:

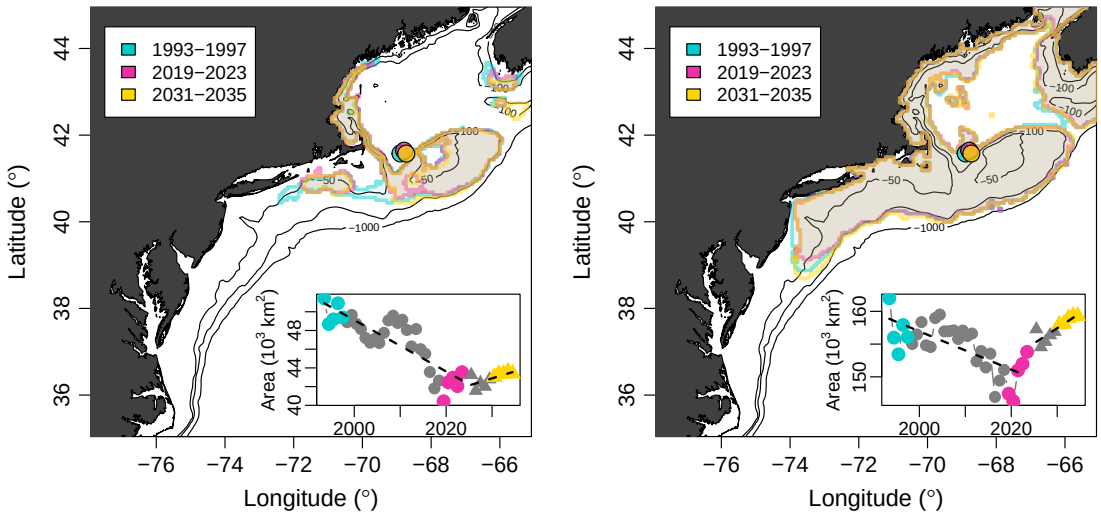


Figure 7: Distribution shifts on the Northeast Shelf for Yellowtail flounder. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Yellowtail flounder core habitat and range decreased significantly between 1993 and 2023. Core habitat loss occurred in southern New England and along the coast of the Gulf of Maine. Yellowtail flounder

lost habitat at the southern edge of their range in the Mid Atlantic Bight (Figure 7). The distribution centroid shifted 11 km east-northeast.

Under near future ocean conditions, yellowtail flounder are expected to gain core habitat and ranges are expected to expand. Yellowtail flounder are predicted to gain more core habitat over Georges Bank. Ranges are predicted to expand southward in the Mid Atlantic Bight (Figure 7). The distribution centroid is predicted to shift 9 km southeast.

In the previous assessment, yellowtail flounder received a High Distributional Vulnerability Rank score due to high larval dispersal, moderate mobility, and variety of preferred habitats (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	50,189	157,664	3.68	7.22
2019-2023	11.2 km ENE	42,445 (-7,744)	150,197 (-7,467)	3.42 (-0.25)	6.81 (-0.40)
2031-2035	8.8 km SE	43,367 (+921)	158,740 (+8,543)	3.55 (+0.12)	6.98 (+0.16)

Directional Effect:

Experts predicted that the effects of climate change on Yellowtail flounder would be **negative** for both the Georges Bank and Southern New England/Mid-Atlantic stocks. Certainty for both stocks was high at 98% and 100% respectively. In the 2016 Climate Vulnerability Assessment, experts also predicted that yellowtail would be negatively impacted by climate change (Hare et al. 2016). For the Gulf of Maine/Cape Cod stock, the effect was predicted to be **neutral**, but with lower certainty (86%). Increased temperatures have been linked to decreased productivity, increased larval mortality, and reduced settlement rates (J. A. Kittel et al. 2026) (Klein, Smith, and Kritzer 2017). Recruitment has also been linked to the Mid-Atlantic Cold Pool (J. Kittel 2025).

Climate Effects on Abundance and Distribution

Yellowtail flounder abundance, distribution, and recruitment have been associated with a number of environmental and climate indices. Rapid warming of the Northwest Atlantic has resulted in decreased productivity in yellow flounder (J. A. Kittel et al. 2026). Higher temperatures can increase mortality in yellowtail flounder larvae, and faster rates of warming causing earlier stratification may reduce settlement rates in juveniles (Klein, Smith, and Kritzer 2017). Yellowtail recruitment in Southern New England is highly correlated with the North Atlantic Oscillation (NAO) and air temperature (Sullivan, Cowen, and Steves 2005). Low recruitment since 1990 was associated with unfavorable environmental conditions caused by a northward shift in the Gulf Stream (Xu 2016) (J. Kittel 2025). The Mid-Atlantic Cold Pool, bottom temperature, and the Atlantic Multidecadal Oscillation (AMO) are also linked to recruitment (J. Kittel 2025). As warming continues, joint management between the United States and Canada may be more challenging as distributions change (Sumaila, Palacios-Abrantes, and Cheung 2020).

Life History Synopsis

Yellowtail flounder are found in the Northwest Atlantic from Newfoundland to Chesapeake Bay, inhabiting sandy bottom waters between 130-230 m depth (Johnson et al. 1999) and at temperatures of -1 to 18°C (Overholtz and Cadrin 1998). Previous research has suggested that fish live up to 17 years, possibly as high as 20 years, although fish older than 7 years is unusual (Northeast Fisheries Science Center (NEFSC) 2012). The fishery is divided into 3 stocks: Gulf of Maine/Cape Cod, Georges Bank, and Southern New England/Mid-Atlantic (Northeast Fisheries Science Center (NEFSC) 2025a) (Northeast Fisheries Science Center (NEFSC) 2025b) (Northeast Fisheries Science Center (NEFSC) 2025c). Most females are mature by ages 2-3 (Northeast Fisheries Science Center (NEFSC) 2012). Spawning occurs from mid-February to July along the continental shelf waters of Georges Bank, northwest of Cape Cod, Southern New England and nearshore along New Jersey and southern Long Island (Martin and Drewry 1978). Spawning peaks from April to June in southern New England and late June to July in Canada (Berrien and Sibunka 1999). Laboratory studies show eggs begin to hatch 6-7 days after fertilization (Benoit and Pepin 1999). At sea hatching times range from 14.5 days at 4 C to 4.5 days at 14 C (Walsh and Morgan 2004). Eggs are pelagic (most at 10-90m depth), with movement limited to currents in spawning locations (Smith, Sibunka, and Wells 1975). Larvae move from the surface down to about 20 m, with larger larvae making longer migrations. This habit may assist in the dispersal of larvae because circulation at the surface transports the larvae further and is likely associated with feeding behavior (Smith, Sibunka, and Wells 1978). Larvae feed on zooplankton, particularly *Calanus* (J. Kittel 2025). Once larvae reach metamorphosis size (11.6-16 mm) bottom temperature (4-8°C) and depth (41-70m) are the most important parameters for determining settlement (Collette and Klein-MacPhee 2002). Juveniles utilize marine and estuarine sandy and muddy substrates, as well as estuarine and marine water columns (Collette and Klein-MacPhee 2002). Rotifers, copepods, and brine shrimp nauplii were fed to juveniles in laboratory studies (Benoit and Pepin 1999). *Calanus* copepods are considered a critical food source for larvae/juveniles (Purchase, Boyce, and Brown 2000). Amphipods (41% by weight) and polychaetes (38.5%) dominate the diet of adult yellowtail, but they also can consume cnidarians, small crabs and shrimps, mysids, cumaceans, isopods, univalve and bivalve mollusks, echinoderms, and sometimes small fish (Bowman et al. 1987) (Bowman et al. 2000). Similar to other groundfish, yellowtail may be impacted by increasing grey seal and dogfish populations and resulting increased predation pressures (J. Kittel 2025). Other predators of juvenile and adult yellowtail include spiny dogfish, skates, cod, hakes, and monkfish (Johnson et al. 1999). Minimal migration patterns show that Southern New England yellowtail traveled eastward during spring and summer and westward in the fall and winter as a result of seasonal changes in temperature (Smith, Sibunka, and Wells 1975). Recent tagging studies, however, show that individuals generally don't move between stocks (Goethel, Legault, and Cadrin 2015). Yellowtail flounder are managed by the New England Fishery Management Council through the Northeast Multispecies (Groundfish) Fishery Management Plan. The Georges Bank stock is jointly managed by the Council and Canada, as the stock extends beyond the international boundary. The Cape Cod/Gulf of Maine stock is not overfished and not subject to overfishing (Northeast Fisheries Science Center (NEFSC) 2025a). The Georges Bank stock is overfished but under a rebuilding plan; overfishing status is unknown (Northeast Fisheries Science Center (NEFSC) 2025b). The Southern New England/Mid-Atlantic stock is overfished but under a rebuilding plan, but not subject to overfishing (Northeast Fisheries Science Center (NEFSC) 2025c).

Supplemental Scores

Exposure Timeseries

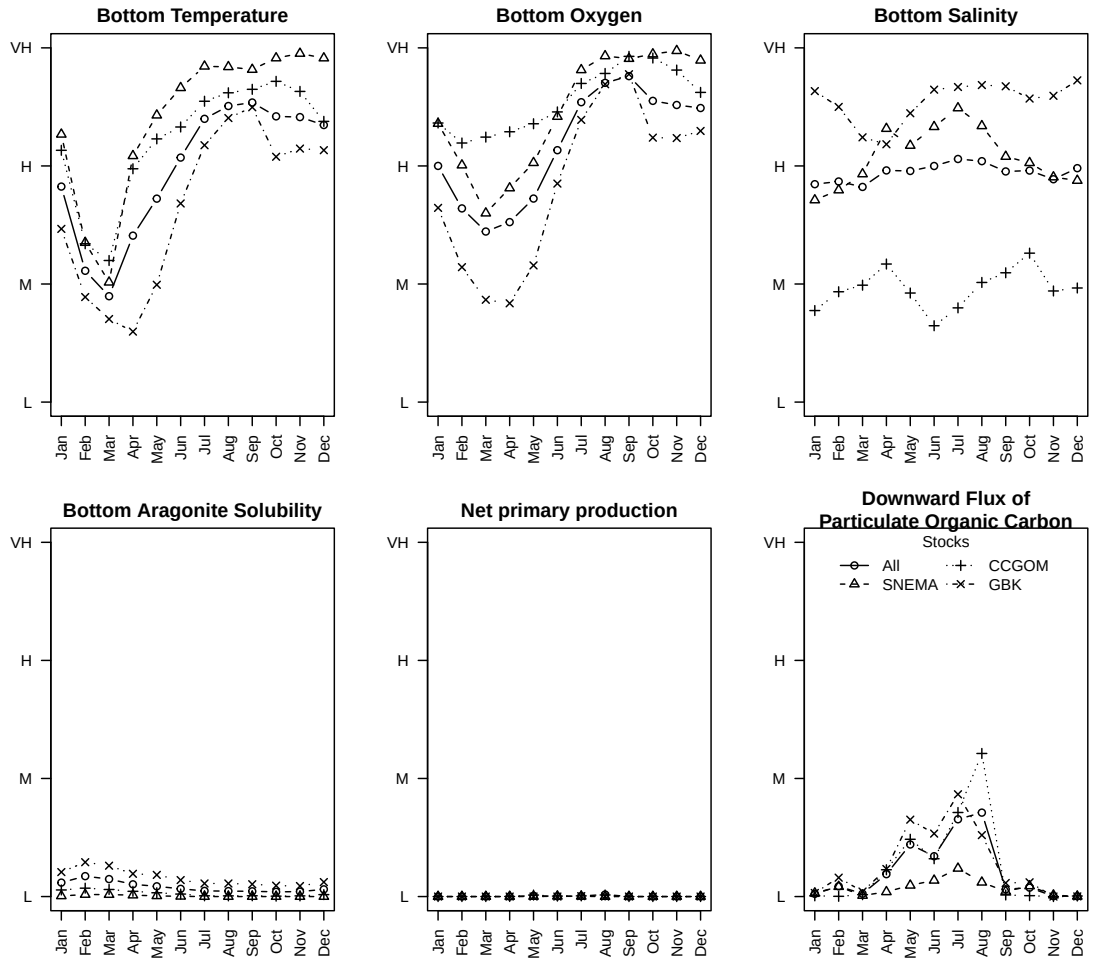


Figure 8: Monthly Timeseries of each Exposure Factor for Yellowtail flounder. Exposure was calculated between 2014-2023 and 2025-2035

Across months, exposure scores ranged from Low to Moderate. Exposure was generally Low in the late winter and early spring (February - May) and High from June through the end of the year (Figure 9). This was driven by bottom temperature and oxygen exposure changes throughout the year (Figure 8).

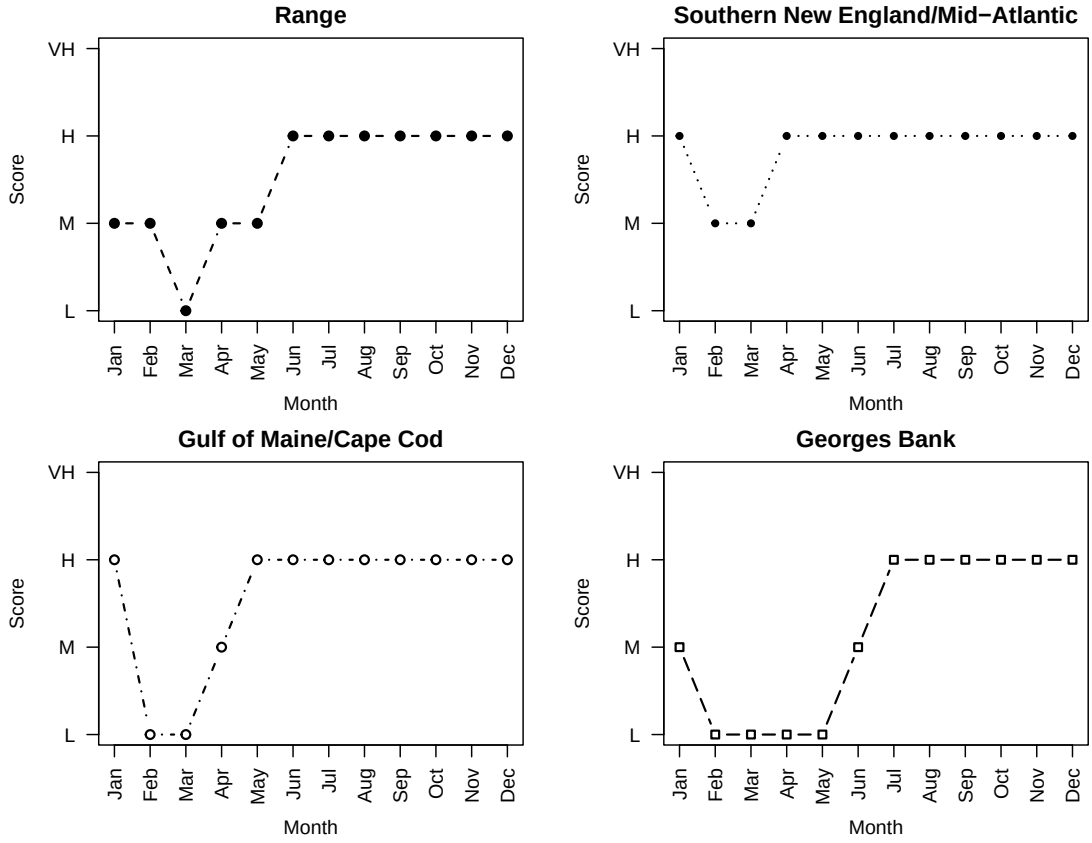


Figure 9: Timeseries of Total Exposure for Yellowtail flounder

Exposure due to Significant Factors

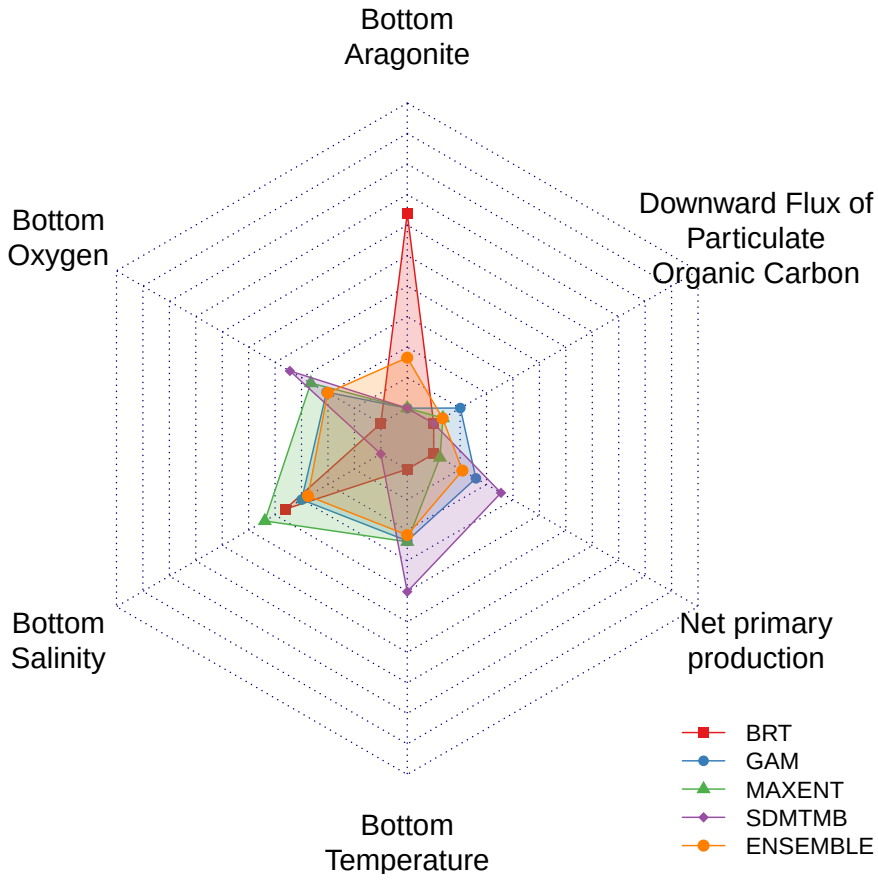


Figure 10: Dynamic Factors driving Yellowtail flounder distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, salinity, oxygen, and aragonite solubility, in addition to net primary productivity were important dynamic drivers of yellowtail flounder species distribution (Figure 10). Yellowtail occupy cooler bottom waters on the continental shelf (Overholtz and Cadrin 1998). Adults primarily feed on amphipods and polychaetes but are generalists (Bowman et al. 1987) (Bowman et al. 2000). When only these factors were used to calculate exposure, mean scores and overall spatial distribution of scores did not change (Figure 12, Figure 11).

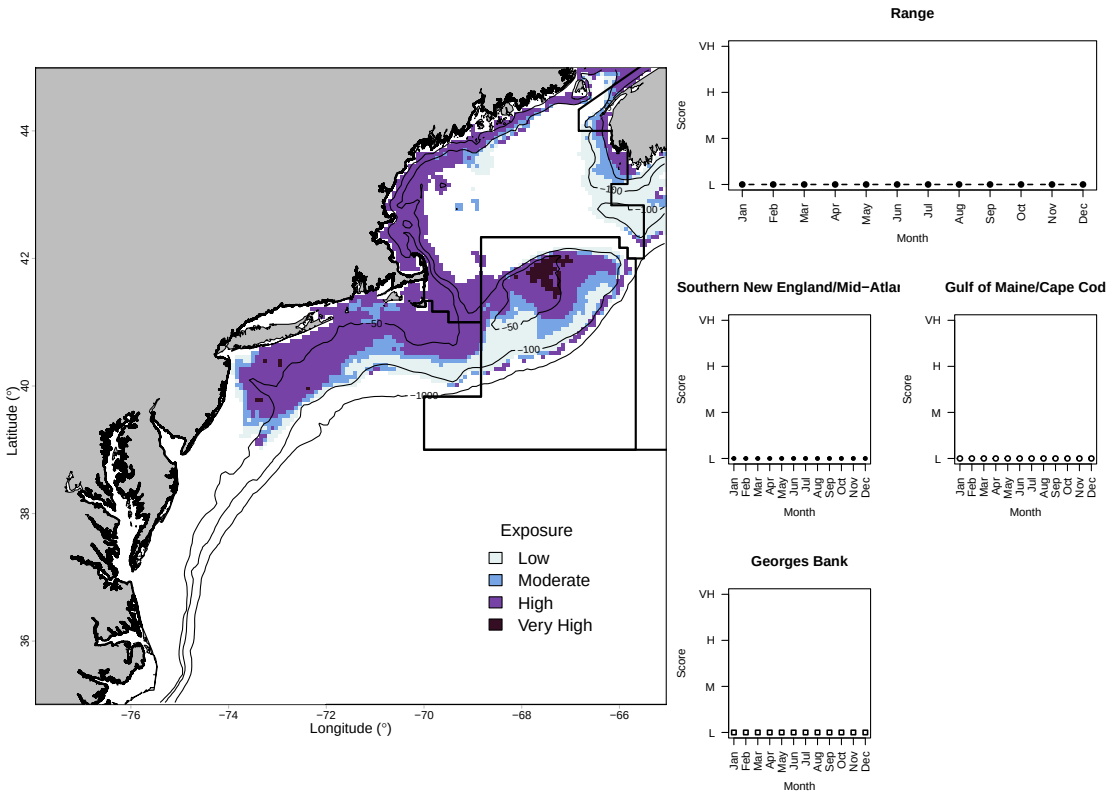


Figure 11: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Yellowtail flounder distributions

Sensitivity

Sensitivity Attributes	Range Score	Range LH DQ	Range Tally	Georges Bank Score	Georges Bank LH DQ	Georges Bank Tally	Gulf of Maine/Cape Cod Score	Gulf of Maine/Cape Cod LH DQ	Gulf of Maine/Cape Cod Tally	Southern New England/Mid-Atlantic Score	Southern New England/Mid-Atlantic LH DQ	Southern New England/Mid-Atlantic Tally
Adult Mobility	2.37	2.36		2.5	2.36		2.32	2.36		2.32	2.36	
Complexity in Reproductive Strategy	1.76	1.97		1.76	2.04		1.76	2.04		1.76	1.84	
Dispersal of Early Life Stages	1.32	2.74		1.32	2.74		1.32	2.74		1.32	2.74	
Egg and Larval Survival and Settlement Requirements	2.57	2.47		2.6	2.5		2.48	2.4		2.64	2.5	
Historical Variability in Temperature	1.57	2.8		1.56	2.8		1.6	2.8		1.56	2.8	
Juvenile Habitat Requirements	1.2	2.5		1.2	2.5		1.2	2.5		1.2	2.5	
Other Stressors	2.29	2.61		2.28	2.6		2.4	2.64		2.2	2.6	
Population Growth Rate	2.11	2.72		2.08	2.72		2.04	2.72		2.2	2.72	
Prey Specificity	1.52	2.7		1.52	2.7		1.52	2.7		1.52	2.7	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.4	2.58		1.4	2.58		1.4	2.58		1.4	2.58	
Spawning Cycle	2.55	2.48		2.56	2.48		2.36	2.48		2.72	2.48	
Stock Size/Status	3.44	2.46		3.56	2.21		2.8	2.75		3.96	2.43	
Total Sensitivity | Certainty	**Moderate (2) | 0.84**			**Moderate (2) | 0.88**			**Low (1) | 0.56**			**Moderate (2) | 0.88**		

Exposure

Exposure Factors	Range Mean	Range SD	Range Tally	Georges Bank Mean	Georges Bank SD	Georges Bank Tally	Gulf of Maine/Cape Cod Mean	Gulf of Maine/Cape Cod SD	Gulf of Maine/Cape Cod Tally	Southern New England/Mid-Atlantic Mean	Southern New England/Mid-Atlantic SD	Southern New England/Mid-Atlantic Tally
Bottom Temperature	2.8	1.01		2.49	0.95		3	1.02		3.07	0.85	
Bottom Oxygen	3.09	0.89		2.72	0.80		3.46	0.80		3.22	0.81	
Bottom Salinity	2.83	0.91		3.44	0.66		1.97	0.90		3.08	0.43	
Bottom Aragonite Solubility	1.12	0.46		1.29	0.76		1.04	0.23		1.02	0.17	
Net primary production	1	0.03		1	0.01		1	0.03		1	0.02	
Total Exposure	**Moderate (2.34)**	**0.89**		**Moderate (2.21)**	**0.99**		**Moderate (2.44)**	**0.83**		**High (2.5)**	**0.77**	

Vulnerability

Vulnerability	Range Mean	Range SD	Range Tally	Georges Bank Mean	Georges Bank SD	Georges Bank Tally	Gulf of Maine/Cape Cod Mean	Gulf of Maine/Cape Cod SD	Gulf of Maine/Cape Cod Tally	Southern New England/Mid-Atlantic Mean	Southern New England/Mid-Atlantic SD	Southern New England/Mid-Atlantic Tally
Total Vulnerability	**Moderate (3.96)**	**1.88**		**Moderate (4.42)**	**1.98**		**Low (2.5)**	**0.95**		**Moderate (5)**	**1.55**	

Low Moderate High Very High

Figure 12: As in Figure 1 for Yellowtail flounder, but with only significant Exposure factors and the resulting Vulnerability Scores.

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